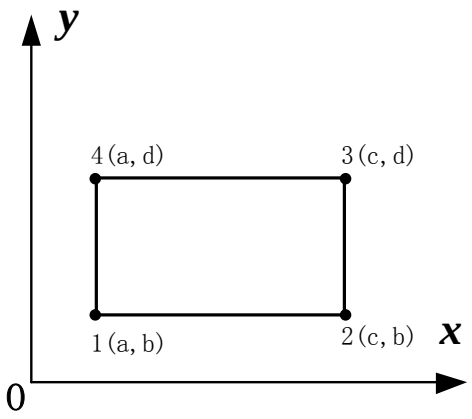
第九讲作业：

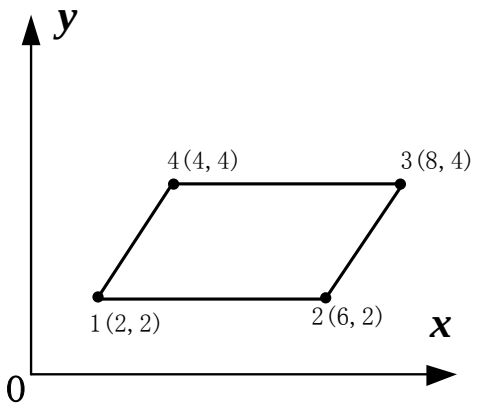
一、图示为四结点四边形等参数单元的子单元，结点坐标在括号内标出。试写出：

1、单元的几何模式；

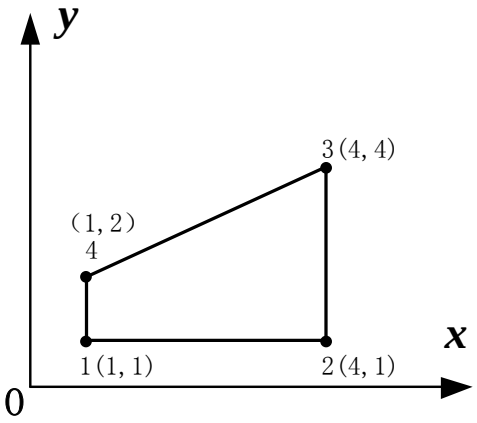
2、单元的Jacobian矩阵 $[J]$和$|J|$，并讨论是否满足等参变换的必要条件。



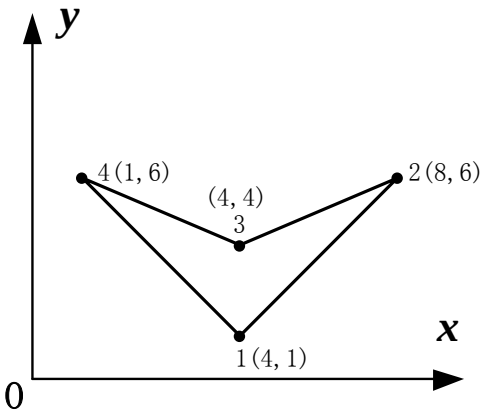
（1）



（2）



（3）

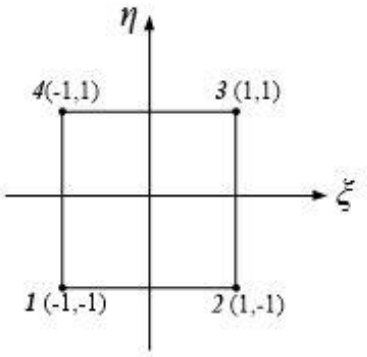


（4）

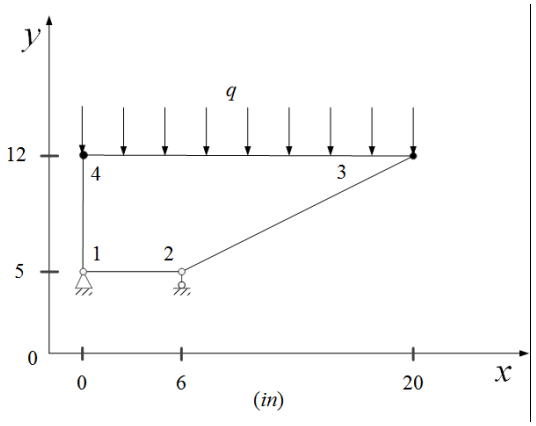
二、四结点等参数单元如图所示，子单元厚度$t = 2\ in$, 3-4 边受面力 $q = 150\ lb/in^2$ 试计算：

1、应变矩阵 $[B(\xi,\eta)]$；

2、等效结点载荷。



母单元（主单元）



子单元（实际单元）

三、

Develop an appropriate mapping to map the quadrilateral shown in Figure 6.42 to the 2×2 square master area. Compute the determinant of the Jacobian matrix and using it demonstrate that the mapping is good. The coordinates of the key points are as follows.

	x	y
1	8.	0.
2	5.65685	5.65685
3	0.	8.
4	0	3
5	2	0

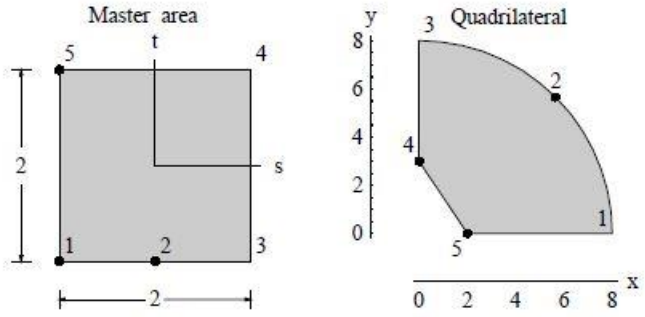


Figure 6.42.